TRƯỜNG ĐẠI HỌC FPT - FPT UNIVERSITY

DEPARTMENT OF INFORMATION TECHNOLOGY
COURSE: PRJ301 - JAVA WEB APPLICATION DEVELOPMENT

Requirement & Design Specification

FPT Classroom Asset Management System (F-CAMS Simple)

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Group: FCAMS Project Team - Group 6
Class: PRJ301 - SE / IA
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I. Overview

The FPT Classroom Asset Management System (F-CAMS Simple) is a specialized Java Web application designed for managing classroom facilities, inventory equipment, and room assets across the FPT University Hoa Lac campus. The project is developed using Java 17 LTS, Jakarta EE 10 (Servlet 6.0, JSP, JSTL 2.0, Custom JSP Tag Library), running on Apache Tomcat 10.1 web container, with data persistence managed by Microsoft SQL Server 2022 via Microsoft JDBC Driver.

The primary objective of F-CAMS Simple is to streamline the operations of campus administrators and IT facility officers in monitoring classroom capacity, tracking asset lifecycle (projectors, air conditioners, Smart TVs, sound equipment, lab workstations), overseeing maintenance status, and preventing asset loss during room reallocations across campus blocks (Alpha, Beta, Delta, Epsilon, Gamma).

1. User Requirements

1.1 Actors

#	Actor	Description
1	Admin	System Administrator possessing full management authority across the entire platform. Has complete CRUD privileges on assets (Add, Edit, Relocate, and DELETE), full management privileges on rooms (Add room with default assets, Edit specifications, and DELETE room), dashboard statistical viewing, and system data reset authority.
2	Staff	Operational Staff / IT Support Officer responsible for daily classroom inspection and equipment handling. Has permissions to view Dashboard KPIs, manage assets (search, filter, add new asset, edit asset info, transfer assets between rooms), and inspect detailed room equipment catalogs. Explicitly RESTRICTED from dangerous operations (cannot delete assets, cannot add, edit, or delete classroom profiles).

1.2 Use Cases

a. Diagram(s)

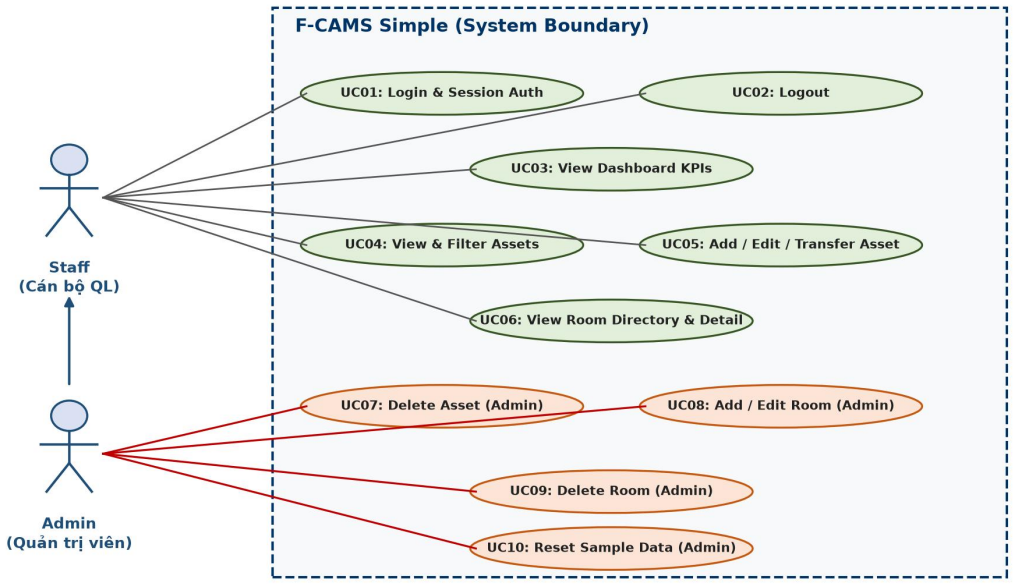


Figure 1.1: Use Case Diagram for F-CAMS Simple

b. Descriptions

ID	Feature	Use Case	Use Case Description
01	Authentication	Login	Users provide username and password to authenticate; stores User in HttpSession.
02	Authentication	Logout	Clears the active HttpSession and redirects user back to the login screen.
03	Dashboard	View Overall KPIs	Displays summary KPI metric cards: Total Assets, Total Rooms, and Total Estimated Inventory Valuation.
04	Dashboard	View Status Distribution	Analyzes operational proportion of assets (Active, Maintenance/Repairing, Faulty/Broken).
05	Asset Management	Search & Filter Assets	Multi-criteria asset filtering by Room, Category, Status, and keyword search.
06	Asset Management	Add & Edit Asset	Registers new asset or updates specifications, serial, unit price, status, notes.
07	Asset Management	Transfer Asset	Relocates asset to another classroom and automatically records historical movement notes.
08	Asset Management	Delete Asset	Permanently deletes an asset record from database (Admin only; enforced by AuthFilter).
09	Room Management	View & Filter Rooms	Lists campus rooms with building filter, chair variance/issue filter, and search.
10	Room Management	Add / Edit / Delete Room	Admin manages room specs (floor, seats, purpose) and associated assets.

2. Overall Functionalities

2.1 Screens Flow

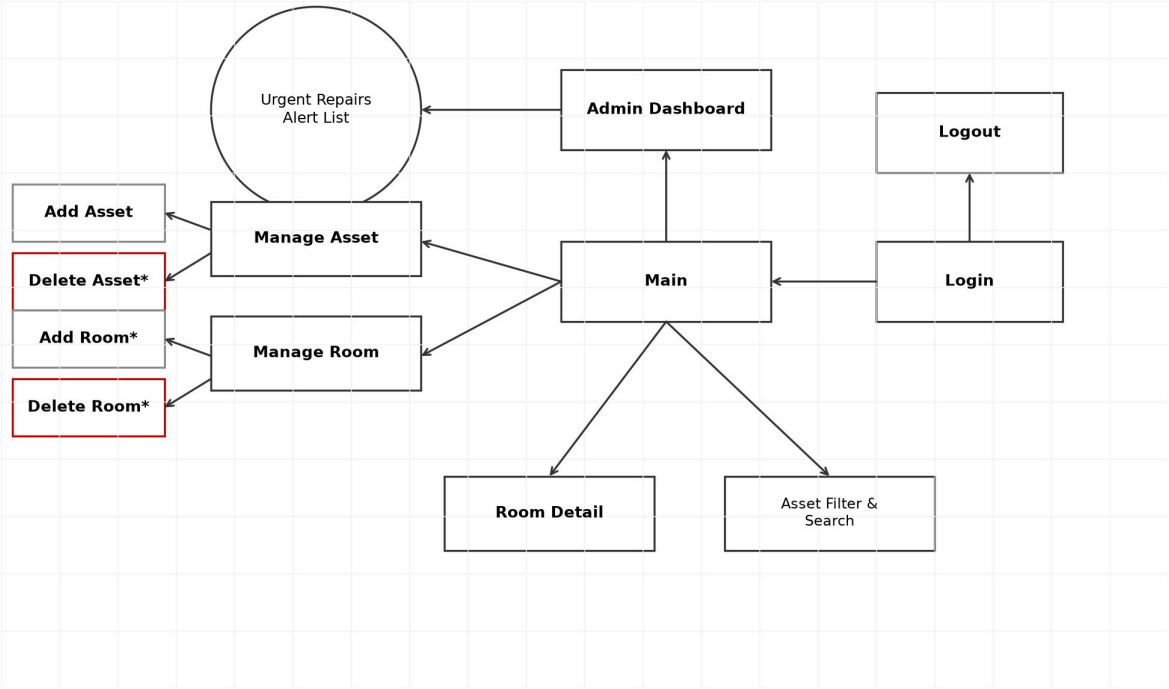


Figure 2.1: Screens Flow Diagram

2.2 Screen Descriptions

#	Feature	Screen	Description
1	Authentication	Login Screen	Provides username and password entry form with validation alerts and role authentication.
2	Common UI	Header Navigation	Displays current user credentials, RoleName custom tag, global navbar tabs, and alert message banners.
3	Dashboard	Dashboard Screen	Presents overall KPI cards (assets, rooms, valuation), asset operational health distribution table, recent additions, and urgent maintenance lists.
4	Asset Management	Asset Management Screen	Centralized inventory control screen with multi-select filter toolbar, search bar, Add/Edit/Transfer forms, and tabular asset data.
5	Room Management	Room Management Screen	Classroom directory showing building, floor, chair variance (ChairStatus custom tag), core equipment indicators, and Add/Edit/Delete room controls.
6	Room Detail	Room Detail Screen	Detailed classroom profile showing capacity specs, educational purpose, and complete listing of all physical assets stationed in the room.

2.3 Screen Authorization

#	Screen / Action	Admin	Staff
1	Login Screen (/login)	X	X
2	Logout (/logout)	X	X
3	Dashboard Screen (/dashboard)	X	X
4	Asset Management - View & Filter (/assets)	X	X
5	Asset Management - Add / Edit / Transfer Asset	X	X

6	Asset Management - DELETE Asset (Dangerous)	X	
7	Room Management - View & Filter Rooms (/rooms)	X	X
8	Room Management - Add / Edit / DELETE Room (Dangerous)	X	
9	Room Detail Screen (/room-detail)	X	X

2.4 Database Design

a. Database Schema

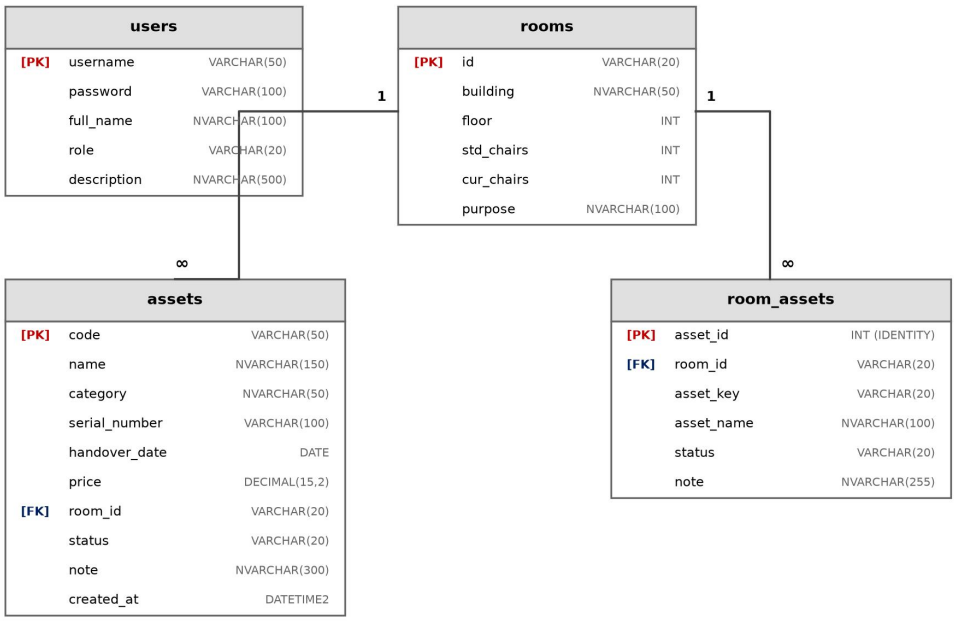


Figure 2.2: Entity Relationship Diagram (ERD) - FCAMS_DB

b. Table Descriptions

No	Table	Description	Keys & Constraints
01	users	Stores system account credentials, full names, authorization roles (ADMIN, STAFF, IT_STAFF, LECTURER), and role descriptions.	Primary Key: username Check: role IN ('ADMIN', 'STAFF', 'IT_STAFF', 'LECTURER')
02	rooms	Manages campus classroom spaces, physical building locations, floor numbers, standard seat quotas, current seats, and intended educational purposes.	Primary Key: id Check: building IN ('Alpha', 'Beta', 'Delta', 'Epsilon', 'Gamma'), floor 1..10, std_chairs 10..100
03	assets	Comprehensive inventory registry holding all physical classroom assets, categories, manufacturer serial numbers, acquisition date, unit price, assigned room, and health status.	Primary Key: code Foreign Key: room_id REFERENCES rooms(id) ON DELETE CASCADE

			Check: status IN ('ACTIVE', 'MAINTENANCE', 'BROKEN', 'REPAIRING', 'FAULTY'), price >= 0
04	room_assets	Tracks 4 fixed core technical installations per classroom (projector, acLeft, acRight, tv) for quick facility health monitoring.	Primary Key: asset_id Foreign Key: room_id REFERENCES rooms(id) ON DELETE CASCADE Unique: (room_id, asset_key)

II. Design Specifications

1. Login Screen

Staff and Administrator log in with their designated credentials to access the management portal.

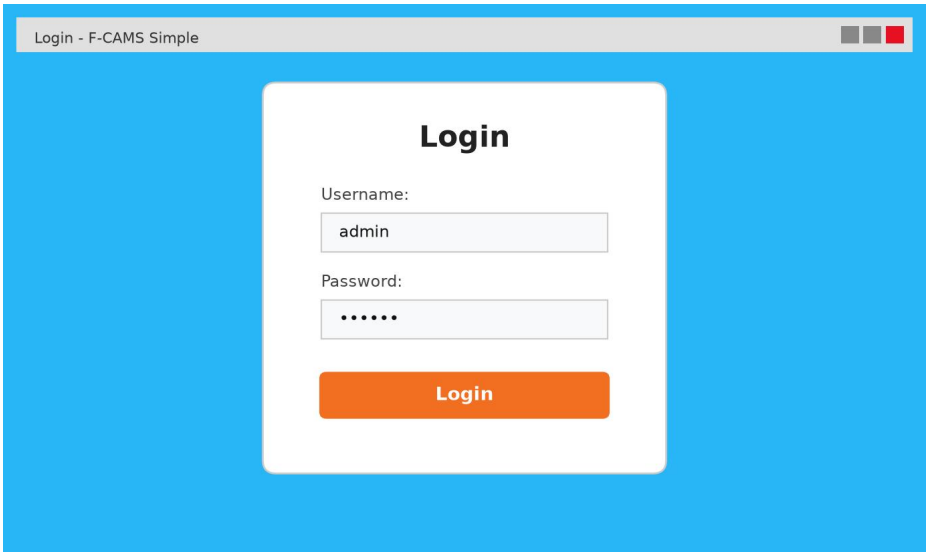


Figure 3.1: Login Screen Interface

Field Name	Field Type	Description
Username	Text Box (Required)	User enters their account username (min 3 characters, e.g. admin, it_beta, hungnq).
Password	Password Box (Required)	User enters their secret authentication password (e.g. 123).
Login	Submit Button	Sends HTTP POST request to /login to authenticate credentials against dbo.users.
Reset	Reset Button	Clears all currently entered values in username and password input fields.

2. Dashboard Screen (Bảng điều khiển Thống kê)

Displays comprehensive high-level system indicators, asset operational health ratios, recently added equipment, and urgent maintenance alerts.

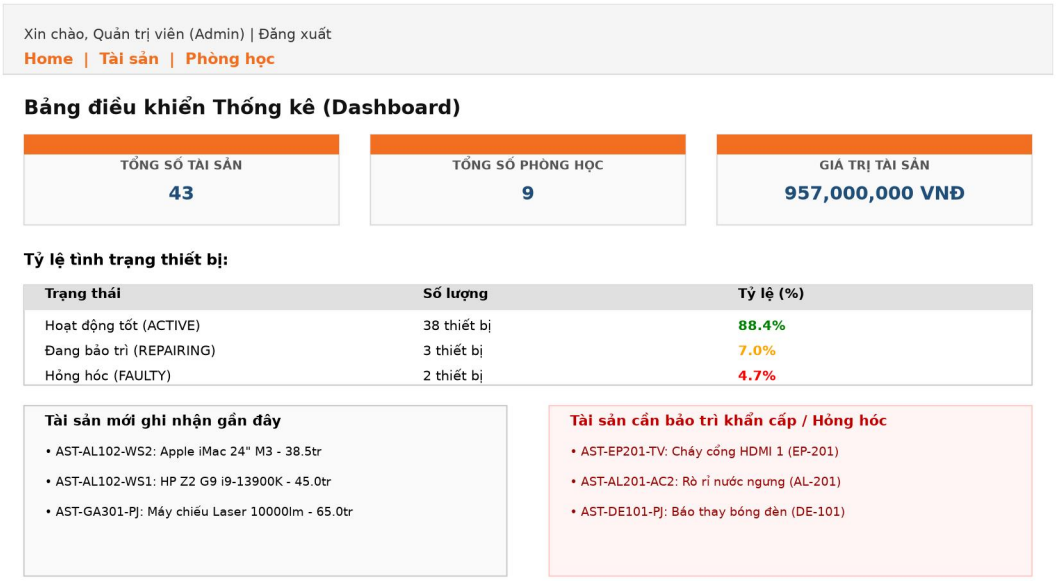


Figure 3.2: Dashboard Interface with KPI Metric Cards

Element / Field Name	Element Type	Description
Total Assets Card	Metric Box (Read-only)	Displays total count of all inventory assets recorded in database.
Total Rooms Card	Metric Box (Read-only)	Displays total number of active classrooms across all campus buildings.
Estimated Value Card	Metric Box (Read-only)	Displays total procurement price sum of all assets, formatted in VND (###,###,### VNĐ).
Asset Health Table	Data Table	Breaks down assets into Active, Repairing, and Faulty counts with calculated percentage ratios.
Recent Assets & Urgent Lists	Data Tables	Displays two responsive tables highlighting newly registered items and urgent maintenance tickets.

3. Asset Management Screen (Quản lý Tài sản - CRUD)

Central management portal for classroom inventory: multi-criteria search and filter, asset registration, asset detail update, inter-room transfer, and deletion.

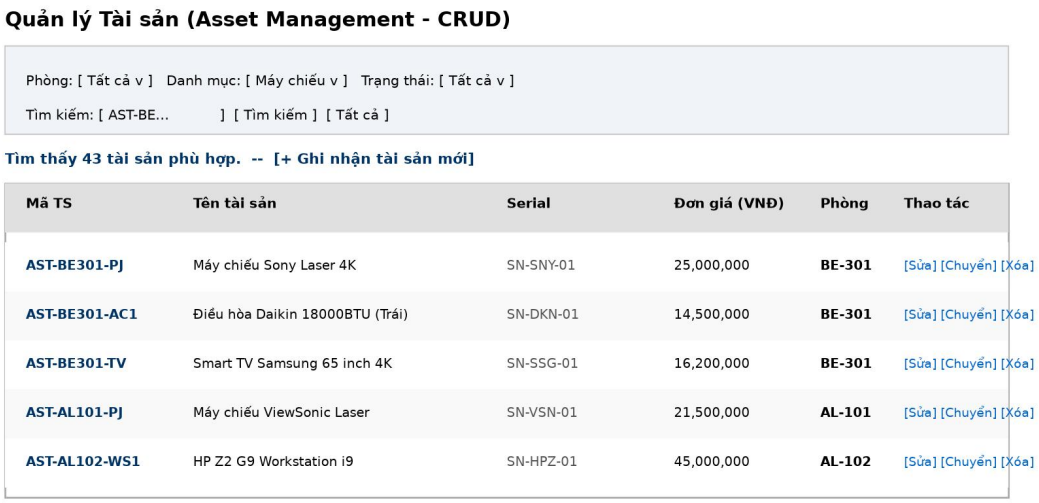


Figure 3.3: Asset Management Screen with Multi-Criteria Filter Bar

Field Name	Field Type	Description
Room Filter	Dropdown List	Filters asset records by assigned classroom (or ALL).
Category Filter	Dropdown List	Filters assets by equipment type (Máy chiếu, Điều hòa, Smart TV, Âm thanh, Thiết bị Lab, Bàn ghế, Khác).
Status Filter	Dropdown List	Filters assets by operational health: ACTIVE (Tốt), REPAIRING (Bảo trì), FAULTY (Hỏng).
Keyword Input	Text Box	Searches matches in asset code, asset name, or manufacturer serial number.
Mã tài sản (code)	Text Box (Required)	Unique asset code identifier (min 3 characters, e.g. AST-BE301-PJ). Primary key.
Tên tài sản (name)	Text Box (Required)	Descriptive asset name including brand and specifications.
Thao tác (Actions)	Buttons / Links	[Sửa] opens inline edit form; [Chuyển phòng] opens relocation form; [Xóa] triggers deletion (Admin only).

4. Room Management Screen (Quản lý Phòng học - CRUD)

Directory of campus classrooms with capacity tracking, chair discrepancy detection via custom tag, and complete room management.



Figure 3.4: Room Management Screen with Chair Quota Variance Tag

Field Name	Field Type	Description
Building Filter	Dropdown List	Filters rooms by campus block (Alpha, Beta, Delta, Epsilon, Gamma, or ALL).
Status Filter	Dropdown List	Filters rooms by condition: ISSUES (seat shortage or broken gear) or OK.
Mã phòng (id)	Text Box (Required)	Room code formatted strictly to regex ^[A-Z]{2}-\d{3}$ (e.g. BE-303, AL-101).
Tầng (floor)	Number Box (1 - 10)	Floor level in building where the classroom is situated.
Ghế (std / cur)	Number Boxes	Standard seating capacity and current physical chair count (triggers chairStatus custom tag).

5. Room Detail Screen (Chi tiết Phòng học & Thiết bị)

Dedicated room inspection screen displaying complete room specifications, seating quota status, and a full catalog of all assets deployed in the room.

<< Quay lại Danh mục phòng học | Xem trên trang Quản lý Tài sản >>

Chi tiết Phòng học: BE-301 (Tòa Beta -- Tầng 3)

- Mục đích sử dụng: Lý thuyết tiêu chuẩn
- Sức chứa / Chỗ ngồi: 35/35 (Đủ) • Tổng số tài sản: 5 tài sản [Sửa thông tin phòng]*
- Thiết bị cố định: Máy chiếu (Tốt), Điều hòa Trái (Tốt), Điều hòa Phải (Tốt), Smart TV (Tốt)

Toàn bộ tài sản và trang thiết bị trang bị bên trong phòng BE-301: [+ Thêm tài sản]

Mã TS	Tên tài sản & Mô tả	Danh mục	Serial	Đơn giá	Tình trạng
AST-BE301-PJ	Máy chiếu Sony Laser 4K	Máy chiếu	SN-SNY-01	25,000,000	Tốt
AST-BE301-AC1	Điều hòa Daikin 18k (Trái)	Điều hòa	SN-DKN-01	14,500,000	Tốt
AST-BE301-AC2	Điều hòa Daikin 18k (Phải)	Điều hòa	SN-DKN-02	14,500,000	Tốt
AST-BE301-TV	Smart TV Samsung 65 inch	Smart TV	SN-SSG-01	16,200,000	Tốt
AST-BE301-AU	Bộ Micro Shure & Amply	Âm thanh	SN-SHR-01	8,500,000	Tốt

Figure 3.5: Room Detail Screen with Equipped Assets Catalog

Element / Field Name	Element Type	Description
Room Profile Summary Card	Information Card	Displays Room ID, Building, Floor, Educational Purpose, and Chair Variance indicator.
Equipped Asset Catalog Table	Data Table	Renders tabular list of all assets deployed in this room, complete with code, name, category, serial, price, status, and notes.
Link: Add Asset to Room	Hyperlink	Navigates directly to /assets?roomId=... with target room pre-selected in registration form.
Item Actions	Action Links / Buttons	Allows inline editing [Sửa], transfer to another room [Chuyển], or deletion [Xóa] (Admin only).